

# GALA: Generation-Aware Cross-Modal Alignment for Text-to-Time-Series Synthesis

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Synthesizing time series from natural language is emerging as the most expressive form of controllable time series generation. However, existing text-conditioned generators either take caption embeddings frozen from off-the-shelf text encoders, or adapt the encoder end-to-end, letting the denoising loss shape the embeddings only as a by-product. In either case, the conditioning representation is never deliberately matched to the signal modality, leaving it ill-suited to guide generation. We address this by introducing GALA: Generation-Aware cross-modal Alignment for text conditional time series generation. GALA is a two-stage approach that first contrastively couples a pretrained text encoder with a time-series foundation model into a shared embedding space with both encoders adapted to generation by an auxiliary generative loss, and then freezes the resulting caption embedding to drive a flow-matching generator. On TSFragment-600K, spanning four domains and three fragment lengths, GALA sets a new state of the art, ranking first in 30 of 36 metric columns and reaching an average rank of 1.08/1.08/1.42 at lengths 24/48/96 against 1.92/2.00/1.75 for the strongest baseline. We further find that generator-internal text encoders force a trade-off between fidelity and caption adherence, whereas conditioning on the aligned embedding breaks it: FID, CTPP, and JFTSD all improve at once. Ablating the auxiliary loss degrades FID, CTPP and JFTSD together, it indicates the generative term is a necessary component of the alignment rather than an add-on.

## 1 Introduction

Time series generation plays an important role in domains such as energy, traffic, and healthcare, where high-quality synthetic series can support data augmentation, privacy-preserving data sharing, and simulation (Yoon et al., 2019; Li et al., 2022). However, the controllability of a generator directly determines its practical value. Unconditional generation methods (Desai et al., 2021; Kong et al., 2020; Yuan & Qiao, 2024; Naiman et al., 2024; Hu et al., 2024) are not controllable. Conditioning on structured signals such as class labels or attribute vectors (Lee et al., 2023; Narasimhan et al., 2024; Shankar et al., 2025; Jing et al., 2024) restricts the user to a predefined condition schema and supports no scenario outside it. In contrast, a natural language description can be both flexible for users and semantically meaningful for describing time series. For example, a clinician can specify a 12-lead ECG through a clinical report (Lai et al., 2025). This makes text the most promising interface for describing the morphology of time series (Lan et al., 2026). Consequently, text-conditional time series generation has emerged as the frontier of conditional time series generation (Ge et al., 2025; Gu et al., 2025; Li et al.,

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2025).

Existing text-conditioned generators differ mainly in where generation is performed. One line of work first trains an autoencoder to compress the time series and then a generator in the latent space (Ge et al., 2025; Lai et al., 2025), while the other line generates the raw series directly in the observation space (Gu et al., 2025; Li et al., 2025). Despite this difference, the two approaches share the same treatment of the text condition and lack a deliberate design of the conditioning path. T2S (Ge et al., 2025), DiffuSETS (Lai et al., 2025), and BRIDGE (Li et al., 2025) obtain fixed caption representations from an off-the-shelf language embedding model and feed them to the generator, and VerbalTS (Gu et al., 2025) learns an adapter that reprograms the frozen text-encoder output into multi-resolution features. In all these methods, the text representation is either kept frozen from a text-only encoder or shaped only by the denoising objective, never explicitly aligned with the time-series modality.

However, explicit cross-modal alignment is effective in other time series tasks. For forecasting, TimesCLIP (Dong et al., 2025) contrastively aligns visual and textual views of the series, BALM-TSF (Zhou et al., 2025b) balances the two modalities with a contrastive semantic-alignment loss, and Time-MMD (Liu et al., 2024) shows that aligned textual context substantially reduces forecasting error; for understanding and reasoning, TEST (Sun et al., 2024) and ChatTS (Xie et al., 2025) align series to an LLM via text prototypes or synthetic captions. Similar observations hold beyond time series (Radford et al., 2021; Ramesh et al., 2022). Converting semantic correspondence into an explicit geometric relation improves downstream performance.

In text conditional time series generation, however, developing a good cross-modal alignment is not a direct plug-in of existing methods. We observe two challenges in our experiments. **1)**: Fine-tuning the text encoders inside the generator improves caption adherence at the cost of fidelity (Table 2), which shows that the generation loss alone is insufficient to shape the shared latent space. **2)**: Existing cross-modal alignment methods optimize the shared space for ranking and retrieval, but do not necessarily retain the fine-grained, reconstruction-relevant detail that a generator needs. We address these two challenges by learning a generation-aware text-time-series shared embedding space (Figure 1), which makes the text embedding faithful enough to drive the generator to better caption adherence without sacrificing fidelity.

In this work, we propose GALA (**Generation-Aware cross-modal Alignment**), a two-stage text-to-time-series generation method. In the first stage, GALA contrastively aligns a pretrained text encoder and a time-series foundation model into a shared embedding space, with a magnitude-aware encoding that preserves the absolute level and scale that captions often refer to. To make the alignment generation-aware, we add an auxiliary generation loss that requires the text embedding to remain sufficient to synthesize its paired series; the encoders themselves are adapted with a low-rank update. The auxiliary loss lifts caption adherence and fidelity together rather than trading them off. It is a necessary component rather than an add-on, and it works for any adaptation that reaches into the backbones, the only failing case being a trainable head on frozen features. In the second stage, the aligned text encoder is frozen and its embedding serves as the condition of a flow-matching generator. In summary, our contributions are threefold:

- We identify an overlooked gap in text-to-time-series generators: the conditioning representation is either frozen text-only features or adapted solely through the generation objective; in neither case is it aligned with the time-series modality, so the conditioning signal is under-optimized for generation.

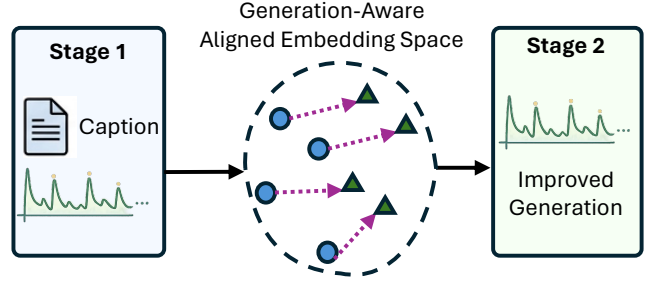


Figure 1: GALA synthesizes a time series from a free-form caption by first aligning text and series into a shared, generation-aware embedding space, then conditioning a generator on the frozen text embedding.

- We propose GALA, a two-stage method that decouples cross-modal alignment from generation and makes the alignment generation-aware: a contrastive objective is combined with an auxiliary generation loss that keeps the text embedding sufficient to synthesize time series.
- We conduct extensive experiments on TSFragment-600K across four domains and three fragment lengths, demonstrating that GALA matches or beats state-of-the-art methods on generation fidelity while improving caption adherence by a large margin. Ablations, a probing analysis, and comparisons with alternative alignment schemes further explain where the gains come from.

## 2 Method

### 2.1 Problem Setup

A caption  $C$  takes values in the caption space  $\mathcal{C}$ , and a time series  $X$  takes values in the time series space  $\mathcal{X} \subset \mathbb{R}^{V \times L}$ , with  $V$  variates and length  $L$ ; the pair  $(C, X)$  is drawn from an unknown joint distribution  $\mu$  with marginals  $\mu_C$  and  $\mu_X$ , of which the training dataset  $\{(c_i, x_i)\}_{i=1}^N$  consists of i.i.d. samples. The aim of text-conditional time series generation is to learn a conditional distribution  $p_\theta(\cdot | c)$ , parameterized by  $\theta$ , that approximates  $\mu(\cdot | c)$  for typical captions, i.e., to solve

$$\min_{\theta} \mathbb{E}_{c \sim \mu_C} [D(\mu(\cdot | c) \| p_\theta(\cdot | c))], \quad (2.1)$$

where  $D$  is any discrepancy measure between distributions. This objective makes precise the two quality axes of our evaluation: **1)** fidelity asks that the generated series look realistic, i.e., the marginal of  $p_\theta$  over captions matches  $\mu_X$ ; **2)** caption adherence asks that each  $p_\theta(\cdot | c)$  put its mass on series whose morphology matches the caption  $c$ , rather than collapsing to the caption-agnostic marginal.

### 2.2 Technical Motivation

Before describing GALA, we give a short information-theoretic argument for its two design decisions: separating alignment from generation, and adding a generative term to the alignment objective. It is not a tight bound or a new theoretical result, but an account of why these two decisions are the natural ones. Anticipating the design, let  $E$  denote the text encoder that maps a caption to its embedding,  $U = E(C)$  that embedding, and  $\mu(\cdot | u)$  the conditional law of  $X$  given  $U=u$  induced by  $\mu$ . GALA fits  $E$  in the first stage, under a contrastive loss  $\mathcal{L}_{\text{align}}$  and an auxiliary generative loss  $\mathcal{L}_{\text{gen}}$  that are defined below, and fits the generator in the second.

**Explicit feature alignment as error decomposition** In GALA, the generator reads the caption through the text encoder:  $p_\theta(\cdot | c) = p_\theta(\cdot | E(c))$ . For any such model, taking  $D$  as KL divergence in the learning objective (2.1) gives the exact identity

$$\begin{aligned} & \underbrace{\mathbb{E}_c[\text{KL}(\mu(\cdot | c) \| p_\theta(\cdot | c))]}_{\text{end-to-end objective}} \\ &= \underbrace{\mathbb{E}_c[\text{KL}(\mu(\cdot | c) \| \mu(\cdot | u))]}_{\mathcal{R}(E): \text{representation error}} + \underbrace{\mathbb{E}_u[\text{KL}(\mu(\cdot | u) \| p_\theta(\cdot | u))]}_{\text{generation error}}, \end{aligned} \quad (2.2)$$

which follows by writing  $\log \frac{\mu(x|c)}{p_\theta(x|c)} = \log \frac{\mu(x|c)}{\mu(x|u)} + \log \frac{\mu(x|u)}{p_\theta(x|u)}$  and taking expectations; the second term reduces to  $\mathbb{E}_u[\dots]$  by the tower property, since  $U = E(C)$  is deterministic and  $X | U \sim \mu(\cdot | u)$ . Under this decomposition,  $\mathcal{R}(E)$  depends only on the encoder; for any encoder, the generation term is minimized separately in Stage 2.

**Alignment needs a generative term for generation** Because  $U = E(C)$  is a deterministic function of  $C$ , the alignment error is exactly a conditional mutual information,  $\mathcal{R}(E) = I(X; C | U) = I(X; C) - I(X; U)$ , where  $I(X; C)$  is a constant of the data; so minimizing  $\mathcal{R}(E)$  is equivalent to maximizing  $I(X; U)$ , and  $\mathcal{R}(E)$  reaches zero iff  $U$  is a sufficient statistic of the caption for the time series. Therefore, the aim of Stage 1 is to make the caption embedding informative about its time series, not merely good at retrieval. This seemingly innocuous difference matters for text-conditional generation. The

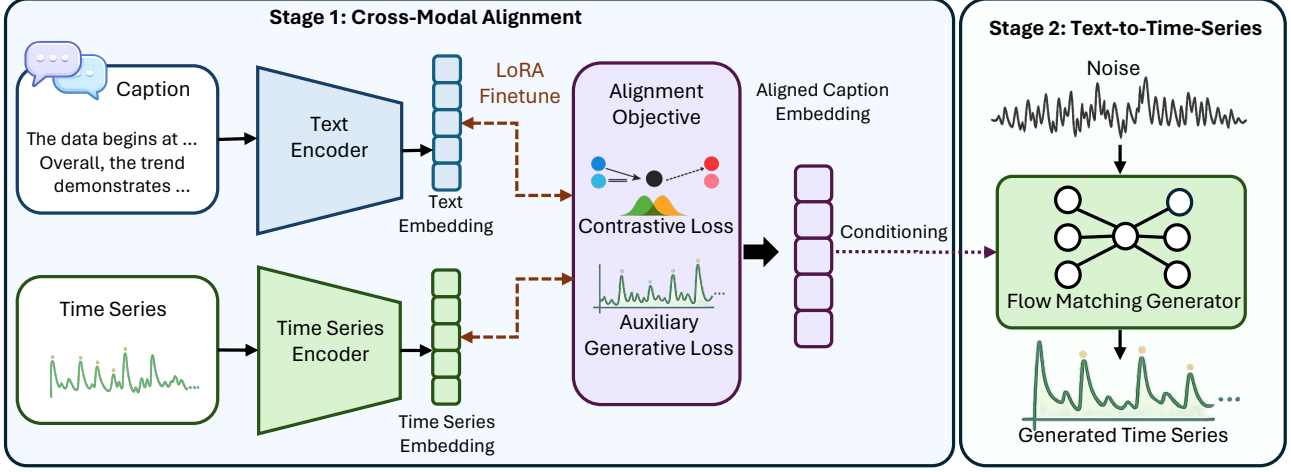


Figure 2: Overview of GALA. Stage 1 contrastively aligns a LoRA-adapted text encoder and time-series encoder into a shared space, with a generation-aware auxiliary loss keeping the text embedding sufficient to synthesize its fragment. Stage 2 freezes the aligned text encoder and trains a flow-matching generator on its embedding.

contrastive loss  $\mathcal{L}_{\text{align}}$  maximizes a lower bound on  $I(X; U)$  (Oord et al., 2018; Poole et al., 2019), namely  $I(X; U) \geq \log B - \mathcal{L}_{\text{align}}$ ; but that bound saturates at  $\log B$ , so once negatives are easily ranked the objective can no longer push for more information. This yields discriminative but not necessarily generation-ready embeddings with only coarse cues about the series.

The generation-aware auxiliary loss adds pressure on the same  $I(X; U)$  from the generative side. Writing the auxiliary diffusion denoiser  $g_{\xi}$  as a conditional density estimate  $q_{\xi}(x | \mathbf{u})$ , its expected log-likelihood is the variational lower bound  $I(X; U) \geq H(X) + \mathbb{E}_{(x, u)} [\log q_{\xi}(x | \mathbf{u})]$ , with  $H(X)$  constant in  $E$ , and  $-\mathcal{L}_{\text{gen}}$  serves as a tractable surrogate for this log-likelihood term. Unlike  $\mathcal{L}_{\text{align}}$ ,  $\mathcal{L}_{\text{gen}}$  keeps rewarding  $\mathbf{u}$  for the finer information about its time series needed for reconstruction, not merely for ranking. Therefore, the auxiliary generation loss delivers better embeddings, lifting text adherence and fidelity together, provided the encoders have the capacity to be reshaped by it, which our ablation (Figure 4) confirms is necessary.

### 2.3 Overview of GALA

GALA generates a time-series fragment from a free-form natural-language description in two decoupled stages (Figure 2). Stage 1 learns a generation-aware shared embedding space in which a time series and its caption are mapped next to one another, by adapting a pretrained text encoder and time-series foundation model. Stage 2 freezes the aligned text encoder and trains a conditional flow-matching generator that turns the text embedding into a fragment. This decoupling is the core design: rather than learning the conditioning representation inside the generator under the reconstruction objective alone, the dedicated alignment stage produces semantically structured conditioning signal for Stage 2 to reuse.

### 2.4 Stage 1: Cross-Modal Alignment

We build a dual encoder (Radford et al., 2021) to map captions and time series onto a shared  $d$ -dimensional unit hypersphere. Captions are encoded with Embedding-Gemma-300M (Vera et al., 2025). The backbone is kept frozen and adapted with LoRA (Hu et al., 2022) on the attention projections  $\{q, k, v, o\}$ . A caption  $c$  is tokenized and passed through the encoder; we mean-pool the last hidden states over non-padding tokens, apply a linear projection  $W_t$ , and  $\ell_2$ -normalize,  $\mathbf{u} = W_t \text{pool}(G_{\phi}(c)) / \|W_t \text{pool}(G_{\phi}(c))\|_2 \in \mathbb{R}^d$ , where  $G_{\phi}$  denotes the LoRA-adapted text encoder. Time series are encoded with Chronos-2 (Ansari et al., 2025), an encoder-only time-series foundation model, again frozen and LoRA-adapted on its attention projections. One additional change is essential: Chronos-2’s instance normalization re-centers and re-scales every series individually, discarding the absolute level and scale that captions often refer

to; we replace it with a dataset-level global  $z$ -score normalization so that magnitude information survives into the embedding (Table 3). The encoded fragment is read out at its [Reg] token at the second-to-last encoder position, averaged over variates, projected by  $W_s$ , and  $\ell_2$ -normalized,  $\mathbf{w} = W_s \text{h}_{\text{glob}}(C_\psi(x)) / \|W_s \text{h}_{\text{glob}}(C_\psi(x))\|_2 \in \mathbb{R}^d$ , with  $C_\psi$  the LoRA-adapted Chronos-2 encoder.

We optimize a contrastive loss plus an auxiliary generation loss to make the two modalities’ features aligned and generation-oriented. **1) Contrastive objective:** For a batch of  $B$  caption–fragment pairs we form the scaled cosine-similarity matrix  $S_{ij} = \mathbf{u}_i^\top \mathbf{w}_j / \tau$  with a fixed temperature  $\tau = 0.07$ . Each caption is paired with a single fragment, so the positives lie on the diagonal and we minimize the standard symmetric InfoNCE loss (Oord et al., 2018),  $\mathcal{L}_{t \rightarrow s} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(S_{ii})}{\sum_{k=1}^B \exp(S_{ik})}$ ,  $\mathcal{L}_{s \rightarrow t} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(S_{ii})}{\sum_{k=1}^B \exp(S_{ki})}$ ,  $\mathcal{L}_{\text{align}} = \frac{1}{2}(\mathcal{L}_{t \rightarrow s} + \mathcal{L}_{s \rightarrow t})$ . **2) Generation-aware auxiliary loss:** Though contrastive alignment makes the text embedding discriminative for time series, it does not guarantee that the embedding retains enough information to steer the generator. We add an auxiliary objective to demand exactly this. A lightweight conditional denoiser  $g_\xi$ , conditioned only on the text embedding  $\mathbf{u}$ , is trained to synthesize the paired time series under the same flow-matching loss used in Stage 2; writing it as  $\mathcal{L}_{\text{gen}}(g_\xi; \mathbf{u}, x)$ , the Stage-1 objective is  $\mathcal{L}_{\text{stage-1}} = \mathcal{L}_{\text{align}} + \lambda_{\text{dw}} \mathbb{E}_{(c,x)}[\mathcal{L}_{\text{gen}}(g_\xi; \mathbf{u}, x)]$ . The denoiser is discarded after Stage 1; its only purpose is to shape the embedding. Crucially, it is conditioned on the *text* side, so a low loss is only achievable when the caption embedding encodes the fragment’s morphology well enough to drive synthesis.

## 2.5 Stage 2: Conditional Generation

Stage 2 trains a conditional generator on top of the frozen aligned text encoder, with the caption embedding  $\mathbf{u}$  as the only conditioning signal. We keep the generator in the observation domain. A time series  $x \in \mathbb{R}^{V \times L}$  is mapped to a square image  $X = \mathcal{D}(x) \in \mathbb{R}^{V \times m \times m}$  by a delay embedding (Naiman et al., 2024), whose  $i$ -th column is the window  $x[i \cdot s : i \cdot s + m]$  (embedding dimension  $m$ , stride  $s$ );  $\mathcal{D}$  is invertible, so a generated image is read back to a series by  $\mathcal{D}^{-1}$ . For  $f_\theta$  we adopt a DiT-style diffusion transformer (Peebles & Xie, 2023) and train it under the JiT recipe (Li & He, 2026), in which the network outputs a clean-signal estimate while the loss is imposed on velocities. Given  $X$ , we draw  $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$  together with a logit-normal timestep  $t \in (0, 1)$  and interpolate the two linearly,  $z_t = tX + (1-t)\epsilon$ ; the velocity that carries  $z_t$  to the data is  $v^* = (X - z_t)/(1-t)$ . Feeding  $z_t$ ,  $t$  and  $\mathbf{u}$  to the network gives  $\hat{X} = f_\theta(z_t, t, \mathbf{u})$ , whose induced velocity  $\hat{v} = (\hat{X} - z_t)/(1-t)$  is matched to  $v^*$  by minimizing  $\mathcal{L}_{\text{gen}} = \mathbb{E}[\|v^* - \hat{v}\|_2^2]$ . Sampling runs the process backwards: a 50-step deterministic Heun solver transports Gaussian noise along the learned velocity field, and  $\mathcal{D}^{-1}$  maps the resulting image back to a time series.

## 3 Experiments

### 3.1 Experimental Setup and Implementation Details

We use TSFragment-600K (Ge et al., 2025), a large-scale benchmark (>600K pairs) purpose-built for text-to-time-series generation, on its four sources (ETTh1s, ETTh1, electricity, traffic) at three lengths (24, 48, 96), each split 90/10 into training and held-out test sets. We compare against VerbalTS (Gu et al., 2025), T2S (Ge et al., 2025), DiffuSETS (Lai et al., 2025), BRIDGE (Li et al., 2025) and Text2Motion (Guo et al., 2022), all run from their official implementations with adaptation for text-conditional generation. Fidelity is measured by FID and caption adherence by CTPP and JFTSD (Lan et al., 2026), all three scored in the embedding space of an evaluation aligner following the protocol of Lan et al. (2026). Its text tower is BGE-large-en-v1.5 (Xiao et al., 2024) and its time series tower is a convolutional encoder trained from scratch, so the evaluator shares no weights with either GALA tower and inherits nothing from Chronos-2. For the caption-generation probe, we adopt the CaTS-Bench metrics (Zhou et al., 2025a). All experiments run on a single NVIDIA RTX PRO 6000 Blackwell GPU; architectures, hyperparameters and code are in the supplements.

### 3.2 Main Results

**GALA achieves new SoTA.** Table 1 compares our method against five existing methods. Our model is best in 30 out of the 36 columns, for a near-perfect average rank of 1.08/1.08/1.42 at  $L = 24/48/96$



GALA: Generation-Aware Cross-Modal Alignment for Text-to-Time-Series Synthesis

| Method                            | ETTh1s       |             |             | ETTm1        |             |             | Electricity  |             |             | Traffic      |             |             | Avg.        |
|-----------------------------------|--------------|-------------|-------------|--------------|-------------|-------------|--------------|-------------|-------------|--------------|-------------|-------------|-------------|
|                                   | FID↓         | CTTP↑       | JFTSD↓      | FID↓         | CTTP↑       | JFTSD↓      | FID↓         | CTTP↑       | JFTSD↓      | FID↓         | CTTP↑       | JFTSD↓      | Rank↓       |
| <i>Length <math>L = 24</math></i> |              |             |             |              |             |             |              |             |             |              |             |             |             |
| DiffuSETS                         | 0.089        | 0.05        | 0.56        | 0.877        | -0.04       | 1.13        | 0.102        | 0.10        | 0.31        | 0.080        | 0.14        | 0.23        | 5.08        |
| T2S                               | 0.078        | 0.07        | 0.54        | 0.254        | 0.11        | 0.53        | 0.166        | 0.09        | 0.37        | 0.403        | 0.07        | 0.56        | 5.17        |
| BRIDGE                            | 0.028        | 0.03        | 0.53        | 0.050        | 0.06        | 0.38        | 0.057        | 0.07        | 0.29        | 0.117        | 0.05        | 0.33        | 4.67        |
| Text2Motion                       | 0.067        | 0.22        | 0.38        | 0.025        | 0.36        | 0.15        | 0.055        | 0.24        | 0.18        | 0.077        | 0.26        | 0.17        | 3.08        |
| VerbalTS                          | <b>0.019</b> | <u>0.43</u> | <u>0.21</u> | <u>0.019</u> | <u>0.52</u> | <u>0.08</u> | <u>0.025</u> | <u>0.40</u> | <u>0.09</u> | <u>0.073</u> | <u>0.35</u> | <u>0.14</u> | <u>1.92</u> |
| <b>GALA</b>                       | <u>0.024</u> | <b>0.71</b> | <b>0.07</b> | <b>0.015</b> | <b>0.64</b> | <b>0.03</b> | <b>0.018</b> | <b>0.52</b> | <b>0.04</b> | <b>0.050</b> | <b>0.45</b> | <b>0.09</b> | <b>1.08</b> |
| <i>Length <math>L = 48</math></i> |              |             |             |              |             |             |              |             |             |              |             |             |             |
| DiffuSETS                         | 0.054        | 0.04        | 0.53        | 0.131        | 0.09        | 0.38        | 0.063        | 0.09        | 0.22        | 0.152        | 0.04        | 0.33        | 4.83        |
| T2S                               | 0.098        | 0.06        | 0.56        | 0.177        | 0.12        | 0.41        | 0.192        | 0.11        | 0.34        | 0.489        | 0.08        | 0.61        | 5.33        |
| BRIDGE                            | 0.035        | 0.03        | 0.52        | 0.037        | 0.06        | 0.32        | <u>0.031</u> | 0.05        | 0.23        | 0.097        | 0.05        | 0.28        | 4.33        |
| Text2Motion                       | 0.094        | 0.16        | 0.46        | 0.042        | 0.30        | 0.16        | 0.068        | 0.22        | 0.16        | 0.089        | 0.25        | 0.16        | 3.42        |
| VerbalTS                          | <u>0.029</u> | <u>0.43</u> | <u>0.22</u> | <u>0.022</u> | <u>0.46</u> | <u>0.08</u> | 0.038        | <u>0.40</u> | <u>0.08</u> | <b>0.062</b> | <u>0.38</u> | <u>0.10</u> | <u>2.00</u> |
| <b>GALA</b>                       | <b>0.022</b> | <b>0.64</b> | <b>0.08</b> | <b>0.011</b> | <b>0.58</b> | <b>0.03</b> | <b>0.020</b> | <b>0.46</b> | <b>0.04</b> | <u>0.069</u> | <b>0.48</b> | <b>0.09</b> | <b>1.08</b> |
| <i>Length <math>L = 96</math></i> |              |             |             |              |             |             |              |             |             |              |             |             |             |
| DiffuSETS                         | 1.187        | 0.00        | 1.41        | 1.410        | 0.01        | 1.52        | 1.017        | 0.00        | 1.13        | 1.258        | 0.02        | 1.35        | 6.00        |
| T2S                               | 0.171        | 0.06        | 0.60        | 0.183        | 0.13        | 0.37        | 0.526        | 0.06        | 0.65        | 0.606        | 0.10        | 0.72        | 4.67        |
| BRIDGE                            | 0.040        | 0.02        | 0.52        | 0.033        | 0.06        | 0.27        | <u>0.046</u> | 0.04        | 0.22        | <u>0.073</u> | 0.09        | 0.25        | 3.67        |
| Text2Motion                       | 0.102        | 0.20        | 0.41        | 0.171        | 0.24        | 0.28        | 0.171        | 0.18        | 0.26        | 0.122        | 0.25        | 0.19        | 3.50        |
| VerbalTS                          | <b>0.024</b> | 0.46        | <u>0.17</u> | <b>0.023</b> | <u>0.45</u> | <u>0.06</u> | 0.053        | <u>0.36</u> | <u>0.09</u> | <b>0.057</b> | <u>0.42</u> | <b>0.08</b> | <u>1.75</u> |
| <b>GALA</b>                       | <u>0.024</u> | <b>0.65</b> | <b>0.07</b> | <u>0.024</u> | <b>0.51</b> | <b>0.04</b> | <b>0.032</b> | <b>0.40</b> | <b>0.06</b> | 0.088        | <b>0.47</b> | <u>0.11</u> | <b>1.42</b> |

Table 1: Comparison with existing text-to-time-series generation methods on TSFragment-600K across four domains and three lengths ( $L \in \{24, 48, 96\}$ ). Each entry is the mean over 3 independent sampling seeds. Per-cell standard deviations are omitted for space and are uniformly small: the largest of all 216 is 0.0165 (DiffuSETS, FID on ETTh1s-96), the largest of GALA’s 36 is 0.0033, and 98% are below 0.01. Significance test is deferred to Technical Supplement. Bold and underline mark the best and second-best per column, ties broken by the unrounded means. Avg. Rank is the mean rank over that length’s 12 columns.

versus 1.92/2.00/1.75 for the strongest baseline (VerbalTS) and  $\geq 3.0$  for all others. This observation is consistent with the benchmark study of Lan et al. (2026), which also identifies VerbalTS as the strongest existing baseline. The advantage is largest on caption adherence: we obtain the highest CTTP in every one of the 12 dataset-length settings, and also the lowest JFTSD in 11 of 12. On fidelity we remain best on 7 of 12 FID columns and competitive elsewhere. Traffic accounts for half of the six columns we do not lead, and is also by far the least diverse source; see the technical supplement. In short, our method matches or beats prior work on fidelity and improves caption adherence by a large margin.

**Not all alignment transfers to generation.** We isolate how Stage 1 shapes the conditioning embedding by swapping in two representative alignment methods with other factors fixed. STRUCTURE (Gröger et al., 2026) freezes both backbones and trains a per-side head, adding a structure-preserving regularizer that ties each aligned feature to its frozen original’s geometry. M3-JEPA (Lei et al., 2025) regresses one modality’s latent from the other with a Mixture-of-Experts predictor, plus an InfoNCE term. Both are purely representational. GALA instead makes the embedding the sole condition of a denoiser that must synthesize the fragment, optimizing it for generative sufficiency. Figure 3 shows the results. On fidelity, GALA obtains the lowest FID on three of the four domains. For caption adherence: GALA attains the highest CTTP on every domain, and the lowest JFTSD on three of four. M3-JEPA beats STRUCTURE on CTTP and JFTSD across all four domains. This shows that objectives forcing the embedding to encode

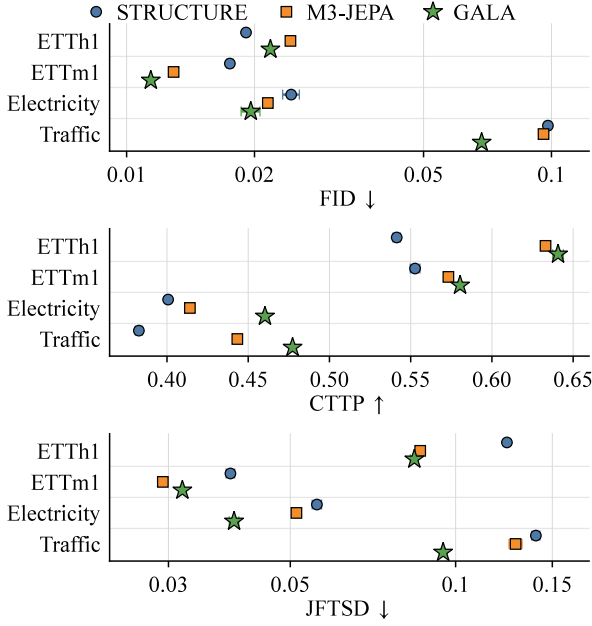


Figure 3: Stage-1 alignment objectives on the four TSFragment-600K domains at length 48, swapping only the alignment stage: STRUCTURE (Gröger et al., 2026), M3-JEPA (Lei et al., 2025), and GALA. Markers are the mean over 3 sampling seeds and error bars the sample std, often narrower than the marker. FID and JFTSD use a logarithmic  $x$ -axis.

| Conditioning             | FID ↓         | CTTP ↑        | JFTSD ↓       |
|--------------------------|---------------|---------------|---------------|
| No-Align (frozen)        | 0.0303        | 0.3169        | 0.2784        |
| No-Align (LoRA)          | 0.0341        | 0.4219        | 0.2207        |
| Align (one-stage)        | 0.0274        | 0.6118        | 0.1098        |
| <b>Align (two-stage)</b> | <b>0.0218</b> | <b>0.6407</b> | <b>0.0841</b> |

Table 2: Effect of the feature-alignment stage on ETTh1s-48. No-Align drops Stage-1 and conditions the generator directly on the frozen or LoRA-tuned text encoder; Align (one-stage) trains aligner and generator in a single run; Align (two-stage) is GALA. Mean over 3 sampling seeds; bold marks the best per column

series morphology better enable the generator to follow the caption.

### 3.3 Ablation Study

**Dedicated alignment stage leads to better generation.** We test the central claim of GALA: a dedicated alignment stage produces a better conditioning signal. The No-Align variants condition the generator on the text encoder directly, frozen or LoRA-tuned; in both, the mean-pooling and linear projector on top are still trained jointly with the generator, so the conditioning representation is shaped only by the generation objective. Align (one-stage) instead trains the full Stage-1 aligner, contrastive loss included, together with the generator. Two observations follow from Table 2: 1) against No-Align (frozen), No-Align (LoRA) improves caption adherence but sacrifices fidelity, so without an alignment stage the two quality axes trade off; 2) Align (one-stage) beats both No-Align variants but trails Align (two-stage), so alignment pays off fully only as a separate stage.

**The auxiliary loss is necessary in Stage 1.** We sweep how the encoders are adapted (LoRA, FT-Full with both backbones trainable, or FT-Head with a trainable head on frozen backbones) against the auxiliary-loss weight  $\lambda_{dw} \in \{0, 1, 2, 4, 8\}$  (Figure 4), where  $\lambda_{dw}=0$  leaves the aligner purely contrastive. The loss is what produces the gain. At its best weight, each architecture that adapts the backbone improves all three metrics at once over  $\lambda_{dw}=0$ : by 3.1/2.9/5.4% (FID/CTTP/JFTSD) for LoRA at  $\lambda_{dw}=1$  and 4.1/6.6/18.2% for FT-Full at  $\lambda_{dw}=2$ , the caption-adherence gains spanning 4 to 25 sampling standard deviations. FT-Head shows that an adaptable backbone is required, since a frozen one leaves the loss nothing to reshape and every non-zero weight is worse than  $\lambda_{dw}=0$ . The sweep also measures sensitivity to  $\lambda_{dw}$ . Each adapting architecture has an interior optimum, at  $\lambda_{dw}=1$  for LoRA and 2 for FT-Full, and raising the weight further gives part of the gain back. We adopt LoRA at  $\lambda_{dw}=1$ , which trains only a small low-rank update instead of both backbones in full.

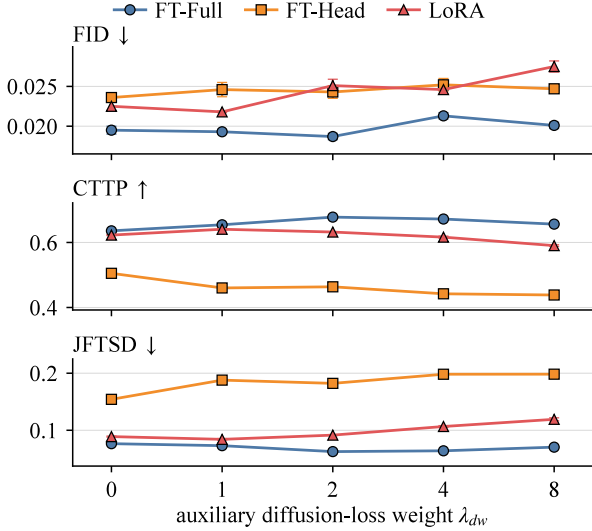


Figure 4: Aligner design sweep on ETTh1s-48: fine-tuning method (FT-Full / FT-Head / LoRA) against the auxiliary-loss weight  $\lambda_{dw} \in \{0, 1, 2, 4, 8\}$  (x-axis, evenly spaced). Markers are the mean over 3 sampling seeds and bars the sample std; LoRA /  $\lambda_{dw}=1$  is our default.

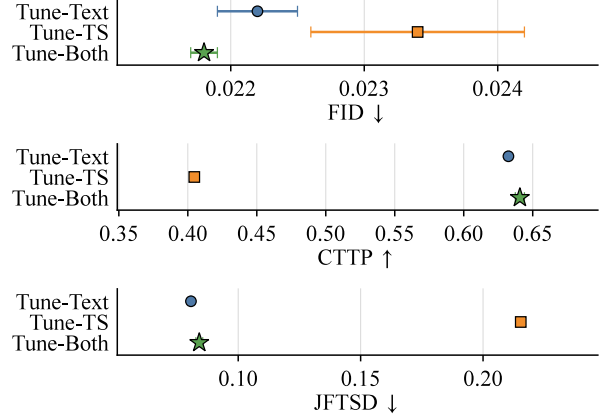


Figure 5: Alignment-side ablation on ETTh1s-48: LoRA-tuning the text tower only, the series tower only, or both (our default) in Stage 1, each frozen and scored with the identical Stage-2 generator. Markers are the mean over 3 sampling seeds; bars are the sample std.

**Tuning the text encoder is what really matters.** We LoRA-tune the text tower only (Tune-Text), the series tower only (Tune-TS), or both (Tune-Both, our default) in Stage 1; on an un-tuned tower the backbone stays frozen and only its single linear projector to the shared space is trained. Figure 5 shows that tuning the text encoder is what really matters: leaving it frozen collapses caption adherence, with Tune-TS at CTPP 0.405 against 0.632 for Tune-Text and 0.641 for Tune-Both, and JFTSD 0.215 against 0.081 and 0.084. Fidelity barely moves, FID spanning only 0.0218–0.0234 across the three variants. Adding the series tower on top of the text tower buys a further gain on CTPP (0.641 vs. 0.632) and FID while costing a little JFTSD, and we adopt Tune-Both as the default.

**Magnitude-preserving normalization carries caption adherence.** Chronos-2 normalizes every fragment individually, erasing the absolute level and scale that captions frequently name; Stage 1 replaces this with a dataset-level global  $z$ -score. We ablate the choice by fitting the aligner under each normalization and training the identical Stage-2 generator on top, on all four domains at length 48. Table 3 shows the effect is concentrated on caption adherence: the global  $z$ -score wins CTPP and JFTSD on every domain, by as much as 0.22 CTPP on ETTm1, whereas FID is close and split (3 of 4 domains in our favour, ETTh1s excepted). Preserving magnitude therefore buys caption adherence at essentially no cost in fidelity.

**The aligned feature better recovers its caption.** Our method targets generation, but the alignment stage also yields a series encoder whose features should carry text-relevant information. As an auxiliary probe, for each encoder variant we LoRA-fine-tune an LLM (Qwen2.5-7B-Instruct) to caption a fragment from its encoded feature and score the captions against the references with CaTS-Bench metrics (Zhou et al., 2025a) (Table 4). To isolate alignment from the magnitude-aware encoding, the Chronos-2 variant is given the same global  $z$ -score normalization, so the two differ only in whether Stage-1 alignment is applied. On the text-overlap metrics all three variants are close, so their coarse semantic content is comparable; they separate on numeric fidelity and min/max recovery, where the aligned feature is clearly best. Since the un-aligned Chronos-2 shares the magnitude-preserving normalization, this gain is attributable to alignment rather than normalization. We read this modestly: alignment does not erase



| Metric  | Norm.      | ETTh1s       | ETTm1        | Elec.        | Traffic      |
|---------|------------|--------------|--------------|--------------|--------------|
| FID ↓   | global     | 0.022        | <b>0.011</b> | <b>0.020</b> | <b>0.069</b> |
|         | per-sample | <b>0.018</b> | 0.018        | 0.024        | 0.087        |
| CTTP ↑  | global     | <b>0.641</b> | <b>0.580</b> | <b>0.460</b> | <b>0.477</b> |
|         | per-sample | 0.546        | 0.358        | 0.431        | 0.432        |
| JFTSD ↓ | global     | <b>0.084</b> | <b>0.032</b> | <b>0.040</b> | <b>0.095</b> |
|         | per-sample | 0.128        | 0.135        | 0.054        | 0.122        |

Table 3: Time-series normalization in the Stage-1 aligner, on all four domains at length 48. global is our dataset-level global  $z$ -score, which preserves absolute level and scale; per-sample is Chronos-2’s native instance normalization. Each aligner is frozen and scored by training the identical Stage-2 generator on top. Mean over 3 sampling seeds; bold marks the better normalization per domain and metric.

| Metric (↑)                                 | Aligned      | Chronos-2 | TSLM         |
|--------------------------------------------|--------------|-----------|--------------|
| <i>Text similarity</i>                     |              |           |              |
| BERT-F1                                    | <b>0.737</b> | 0.735     | 0.736        |
| SimCSE                                     | <b>0.904</b> | 0.889     | 0.893        |
| BLEU                                       | 0.171        | 0.169     | <b>0.175</b> |
| ROUGE-L                                    | <b>0.407</b> | 0.398     | 0.403        |
| METEOR                                     | 0.391        | 0.388     | <b>0.395</b> |
| <i>Numeric &amp; statistical inference</i> |              |           |              |
| Num. Fidelity                              | <b>0.471</b> | 0.302     | 0.360        |
| min-acc                                    | <b>0.545</b> | 0.097     | 0.178        |
| max-acc                                    | <b>0.772</b> | 0.151     | 0.228        |

Table 4: Probing how much textual information the encoded series feature retains, averaged over the four datasets at length 48; bold marks the best per row. Aligned is our cross-modally aligned feature; the pretrained Chronos-2 (Ansari et al., 2025) variant uses the same global  $z$ -score normalization, differing only in alignment; TSLM (Trabelsi et al., 2025) is a further encoder baseline.

text-relevant information and appears to retain level- and range-related cues, but we do not claim it is an optimal captioning feature.

## 4 Related Work

### 4.1 Time Series Generation

Research on time series generation has progressed from unconditional synthesis (Yoon et al., 2019; Desai et al., 2021; Kong et al., 2020; Yuan & Qiao, 2024; Naiman et al., 2024; Hu et al., 2024; Gonen et al., 2025), through conditioning on structured, machine-readable signals (Lan et al., 2026), to conditioning on free-form text. Structured control attaches either a discrete class (TTS-CGAN (Li et al., 2022), TimeVQVAE (Lee et al., 2023)) or low-dimensional attribute metadata (Time Weaver (Narasimhan et al., 2024), WaveStitch (Shankar et al., 2025), TEdit (Jing et al., 2024)) to the generator; it is effective but bounded by a predefined schema.

Free-form text enables flexible conditioning (Lan et al., 2026). Recent generators include DiffuSETS (Lai et al., 2025) for ECG from clinical reports, T2S (Ge et al., 2025), a latent diffusion model with a length-adaptive autoencoder, VerbalTS (Gu et al., 2025), with multi-view denoising for tighter text-signal alignment, and BRIDGE (Li et al., 2025), with multi-agent optimization of text control. In all of them the conditioning path is either a frozen text-only encoder or trained solely under the reconstruction objective, so the representation is never aligned with the series modality. We instead decouple alignment from generation: a dedicated contrastive stage produces a text embedding aligned with the series domain, then frozen and reused by the generator.

### 4.2 Multimodal Feature Alignment

Feature alignment maps heterogeneous modalities into a shared representation, making semantic correspondence a geometric relation; the technique is typically chosen for the downstream task it serves. Contrastive dual-encoder alignment (Radford et al., 2021; Jia et al., 2021; Yu et al., 2022) delivers strong zero-shot classification and cross-modal retrieval, while connector-based methods align a modality to a frozen language model for open-ended understanding, as in Flamingo (Alayrac et al., 2022) and LLaVA (Liu et al., 2023).

Multimodal alignment has also been applied to time series, chiefly with language models (Liu et al., 2025): Time-LLM (Jin et al., 2024) and TimesCLIP (Dong et al., 2025) align series to an LLM representation space for forecasting, and TEST (Sun et al., 2024) and ChatTS (Xie et al., 2025) do so via text prototypes or synthetic captions for question answering and explanation. Our setting is instead generation, where the alignment supplies the conditioning signal.

## 5 Conclusion

We revisited text-to-time-series generation from the standpoint of representation quality and pinpointed a blind spot: by taking the conditioning signal from a text-only encoder or shaping it through the generative loss alone, current methods leave the language–signal correspondence implicit. GALA remedies this with a dedicated alignment stage held apart from synthesis. An information-theoretic reading clarifies why the split is reasonable (the objective factors into a representation and a generation term) and why a contrastive criterion alone falls short (too weak once negatives are easy to rank). Restoring the lost guidance with a generation-aware auxiliary loss, applied to encoders that are themselves adapted, lets both quality facets climb together. Experiments across four domains and three lengths, with ablations, alternative alignment recipes, and a probing study, substantiate it. Future work includes longer signals and other modalities.

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## References

- Alayrac, J.-B., Donahue, J., Luc, P., Miech, A., Barr, I., Hasson, Y., Lenc, K., Mensch, A., Millican, K., Reynolds, M., Ring, R., Rutherford, E., Cabi, S., Han, T., Gong, Z., Samangooei, S., Monteiro, M., Menick, J., Borgeaud, S., Brock, A., Nematzadeh, A., Sharifzadeh, S., Binkowski, M., Barreira, R., Vinyals, O., Zisserman, A., and Simonyan, K. Flamingo: a visual language model for few-shot learning. In *Proceedings of the 36th International Conference on Neural Information Processing Systems, NIPS ’22*. Curran Associates Inc., 2022. ISBN 9781713871088. 9
- Ansari, A. F., Shchur, O., Küken, J., Auer, A., Han, B., Mercado, P., Rangapuram, S. S., Shen, H., Stella, L., Zhang, X., et al. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*, 2025. 4, 9
- Demšar, J. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine learning research*, 7(Jan):1–30, 2006. 19
- Desai, A., Freeman, C., Wang, Z., and Beaver, I. Timevae: A variational auto-encoder for multivariate time series generation. *arXiv preprint arXiv:2111.08095*, 2021. 1, 9
- Dong, S., Fan, W., Wu, T., and Fu, Y. Teaching time series to see and speak: Forecasting with aligned visual and textual perspectives. *arXiv preprint arXiv:2506.24124*, 2025. 2, 10
- Ge, Y., Li, J., Zhao, Y., Wen, H., Li, Z., Qiu, M., Li, H., Jin, M., and Pan, S. T2s: High-resolution time series generation with text-to-series diffusion models. In Kwok, J. (ed.), *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence, IJCAI-25*, pp. 5208–5216. International Joint Conferences on Artificial Intelligence Organization, 8 2025. doi: 10.24963/ijcai.2025/580. URL <https://doi.org/10.24963/ijcai.2025/580>. Main Track. 1, 2, 5, 9, 13, 16
- Gonen, T., Pemper, I., Naiman, I., Berman, N., and Azencot, O. Time series generation under data scarcity: A unified generative modeling approach. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL <https://openreview.net/forum?id=p324ryBKTc>. 9
- Gröger, F., Wen, S., Le, H., and Brbic, M. With limited data for multimodal alignment, let the STRUCTURE guide you. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2026. URL <https://openreview.net/forum?id=IkvQqD7hk3>. 6, 7

- Gu, S., Li, C., Jing, B., and Ren, K. VerbalTS: Generating time series from texts. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=KaukiEQcbk>. 1, 2, 5, 9, 14
- Guo, C., Zou, S., Zuo, X., Wang, S., Ji, W., Li, X., and Cheng, L. Generating diverse and natural 3d human motions from text. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 5152–5161, 2022. 5, 17
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., et al. Lora: Low-rank adaptation of large language models. *Iclr*, 1(2):3, 2022. 4
- Hu, Y., Wang, X., Ding, Z., Wu, L., Zhang, H., Li, S. Z., Wang, S., Zhang, J., Li, Z., and Chen, T. Flowts: Time series generation via rectified flow. *arXiv preprint arXiv:2411.07506*, 2024. 1, 9
- Jia, C., Yang, Y., Xia, Y., Chen, Y.-T., Parekh, Z., Pham, H., Le, Q., Sung, Y.-H., Li, Z., and Duerig, T. Scaling up visual and vision-language representation learning with noisy text supervision. In *International conference on machine learning*, pp. 4904–4916. PMLR, 2021. 9
- Jin, M., Wang, S., Ma, L., Chu, Z., Zhang, J., Shi, X., Chen, P.-Y., Liang, Y., Li, Y.-F., Pan, S., et al. Time-llm: Time series forecasting by reprogramming large language models. In *International conference on learning representations*, volume 2024, pp. 23857–23880, 2024. 10
- Jing, B., Gu, S., Chen, T., Yang, Z., Li, D., He, J., and Ren, K. Towards editing time series. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. URL <https://openreview.net/forum?id=qu5NTwZtxA>. 1, 9
- Kong, Z., Ping, W., Huang, J., Zhao, K., and Catanzaro, B. Diffwave: A versatile diffusion model for audio synthesis. *arXiv preprint arXiv:2009.09761*, 2020. 1, 9
- Lai, Y., Chen, J., Zhao, Q., Zhang, D., Wang, Y., Geng, S., Li, H., and Hong, S. Diffusets: 12-lead ecg generation conditioned on clinical text reports and patient-specific information. *Patterns*, 6(10), 2025. 1, 2, 5, 9, 16
- Lan, S., Gu, S., Xiong, Z., and Ren, K. Contsg-bench: A unified benchmark for conditional time series generation. *arXiv preprint arXiv:2603.04767*, 2026. 1, 5, 6, 9, 17
- Lee, D., Malacarne, S., and Aune, E. Vector quantized time series generation with a bidirectional prior model. *arXiv preprint arXiv:2303.04743*, 2023. 1, 9
- Lei, H., Cheng, X., Qin, Q., Wang, D., Huang, H., Gu, Q., Wu, Y., and Ji, L. M3-JEPA: Multimodal alignment via multi-gate MoE based on the joint-embedding predictive architecture. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*. PMLR, 2025. URL <https://proceedings.mlr.press/v267/lei25b.html>. 6, 7
- Li, H., Huang, Y.-H., Xu, C., Schlegel, V., Jiang, R., Batista-Navarro, R., Nenadic, G., and Bian, J. BRIDGE: Bootstrapping text to control time-series generation via multi-agent iterative optimization and diffusion modeling. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=uRD6wkqulN>. 1, 2, 5, 9, 16
- Li, T. and He, K. Back to basics: Let denoising generative models denoise. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 36115–36125, June 2026. 5
- Li, X., Ngu, A. H. H., and Metsis, V. Tts-cgan: A transformer time-series conditional gan for biosignal data augmentation. *arXiv preprint arXiv:2206.13676*, 2022. 1, 9
- Liu, C., Zhou, S., Xu, Q., Miao, H., Long, C., Li, Z., and Zhao, R. Towards cross-modality modeling for time series analytics: A survey in the llm era. *arXiv preprint arXiv:2505.02583*, 2025. 10
- Liu, H., Li, C., Wu, Q., and Lee, Y. J. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916, 2023. 9
- Liu, H., Xu, S., Zhao, Z., Kong, L., Kamarthi, H., Sasanur, A. B., Sharma, M., Cui, J., Wen, Q., Zhang, C., et al. Time-mmd: Multi-domain multimodal dataset for time series analysis. *Advances in Neural Information Processing Systems*, 37:77888–77933, 2024. 2
- Naiman, I., Berman, N., Pemper, I., Arbiv, I., Fadlon, G., and Azencot, O. Utilizing image transforms and diffusion models for generative modeling of short and long time series. *Advances in Neural Information Processing Systems*, 37:121699–121730, 2024. 1, 5, 9
- Narasimhan, S. S., Agarwal, S., Akcin, O., Sanghavi, S., and Chinchali, S. Time weaver: a conditional

- time series generation model. In *Proceedings of the 41st International Conference on Machine Learning*, ICML'24. JMLR.org, 2024. 1, 9
- Oord, A. v. d., Li, Y., and Vinyals, O. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018. 4, 5
- Peebles, W. and Xie, S. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 4195–4205, 2023. 5
- Poole, B., Ozair, S., Van Den Oord, A., Alemi, A., and Tucker, G. On variational bounds of mutual information. In *International conference on machine learning*, pp. 5171–5180. PMLR, 2019. 4
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pp. 8748–8763. PmLR, 2021. 2, 4, 9
- Ramesh, A., Dhariwal, P., Nichol, A., Chu, C., and Chen, M. Hierarchical text-conditional image generation with clip latents. *arXiv preprint arXiv:2204.06125*, 1(2):3, 2022. 2
- Shankar, A., Chen, L., van Deursen, A., and Hai, R. Wavestitch: Flexible and fast conditional time series generation with diffusion models. *Proc. ACM Manag. Data*, 3(6), December 2025. doi: 10.1145/3769842. URL <https://doi.org/10.1145/3769842>. 1, 9
- Sun, C., Li, H., Li, Y., and Hong, S. Test: Text prototype aligned embedding to activate llm’s ability for time series. In *International Conference on Learning Representations*, volume 2024, pp. 37854–37881, 2024. 2, 10
- Trabelsi, M., Boyd, A., Cao, J., and Uzunalioglu, H. Time series language model for descriptive caption generation. *Engineering Applications of Artificial Intelligence*, 162:112673, 2025. 9
- Vera, H. S., Dua, S., Zhang, B., Salz, D., Mullins, R., Panyam, S. R., Smoot, S., Naim, I., Zou, J., Chen, F., et al. Embeddinggemma: Powerful and lightweight text representations. *arXiv preprint arXiv:2509.20354*, 2025. 4
- Xiao, S., Liu, Z., Zhang, P., Muennighoff, N., Lian, D., and Nie, J.-Y. C-pack: Packed resources for general chinese embeddings. In *Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval*, pp. 641–649, 2024. 5, 17
- Xie, Z., Li, Z., He, X., Xu, L., Wen, X., Zhang, T., Chen, J., Shi, R., and Pei, D. Chatts: Aligning time series with llms via synthetic data for enhanced understanding and reasoning. *Proc. VLDB Endow.*, 18(8):2385–2398, April 2025. ISSN 2150-8097. doi: 10.14778/3742728.3742735. URL <https://doi.org/10.14778/3742728.3742735>. 2, 10
- Yoon, J., Jarrett, D., and Van der Schaar, M. Time-series generative adversarial networks. *Advances in neural information processing systems*, 32, 2019. 1, 9
- Yu, J., Wang, Z., Vasudevan, V., Yeung, L., Seyedhosseini, M., and Wu, Y. Coca: Contrastive captioners are image-text foundation models. *arXiv preprint arXiv:2205.01917*, 2022. 9
- Yuan, X. and Qiao, Y. Diffusion-TS: Interpretable diffusion for general time series generation. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=4h1apFj099>. 1, 9
- Zhou, L., Yashwante, P., Fisher, M., Sampieri, A., Zhou, Z., Galasso, F., and Yu, R. Cats-bench: Can language models describe numeric time series? *arXiv preprint arXiv:2509.20823*, 2025a. 5, 8
- Zhou, S., Schöner, H., Lyu, H., Fouché, E., and Wang, S. Balm-tsf: Balanced multimodal alignment for llm-based time series forecasting. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management*, pp. 4498–4508, 2025b. 2

# Appendix

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This technical appendix provides the full implementation details of GALA that are summarized in the main paper. All values below are the settings used to produce the reported results. Our code and configuration files are released for reproducibility.

## A Datasets and Preprocessing

We use TSFragment-600K (Ge et al., 2025) with four sources (ETTh1s, ETTm1, electricity, traffic) at three fragment lengths  $L \in \{24, 48, 96\}$ . Each (source, length) configuration is split 90/10 into training and held-out test sets. In Stage 1 we replace Chronos-2’s per-instance normalization with a dataset-level global  $z$ -score normalization whose mean and standard deviation are estimated on the training split only, so that absolute level and scale survive into the embedding.

## B Stage-1: Cross-Modal Alignment

Table 5 lists the Stage-1 aligner settings. The auxiliary denoiser is conditioned on the text embedding, trained jointly with the contrastive objective, and discarded after Stage 1.

## C Stage-2: Conditional Generation

Table 6, Table 7, and Table 8 list the Stage-2 generator settings. The generator is conditioned only on the frozen 512-d caption embedding and the timestep.



| Hyperparameter                                      | Value                                                |
|-----------------------------------------------------|------------------------------------------------------|
| <i>Encoders</i>                                     |                                                      |
| Text encoder                                        | Embedding-Gemma-300M (frozen, LoRA)                  |
| Series encoder                                      | Chronos-2 (frozen, LoRA)                             |
| Shared embedding dim $d$                            | 512                                                  |
| Series normalization                                | dataset-level global $z$ -score                      |
| <i>LoRA</i>                                         |                                                      |
| Rank $r$                                            | 64                                                   |
| $\alpha$                                            | 16                                                   |
| Dropout                                             | 0.05                                                 |
| Target modules                                      | $\{q, k, v, o\}$ (both towers)                       |
| Adapted towers                                      | both (text & series)                                 |
| <i>Contrastive objective</i>                        |                                                      |
| Temperature $\tau$                                  | 0.07 (fixed)                                         |
| Loss                                                | symmetric InfoNCE + $\lambda_{dw} \mathcal{L}_{gen}$ |
| $\lambda_{dw}$ (default)                            | 1.0                                                  |
| <i>Optimization</i>                                 |                                                      |
| Optimizer                                           | AdamW ( $\beta=(0.9, 0.999)$ )                       |
| Learning rate                                       | $2 \times 10^{-5}$ (constant, no warmup)             |
| Weight decay                                        | 0                                                    |
| Gradient clipping                                   | 1.0                                                  |
| Batch size                                          | 256                                                  |
| Epochs                                              | 100                                                  |
| <i>Auxiliary denoiser (discarded after Stage 1)</i> |                                                      |
| Backbone                                            | JiT (2-D), hidden 64, depth 2, heads 4               |
| Patch size / image side $m$                         | 2 / 10                                               |
| Timestep schedule                                   | logit-normal ( $P_{mean}=0.5, P_{std}=1.2$ )         |

Table 5: Stage-1 alignment hyperparameters. In the fine-tuning-method ablation, FT-Full uses learning rate  $1 \times 10^{-5}$ ; all other settings are unchanged.

## D Length- and Dataset-Specific Settings

The delay embedding maps a length- $L$  fragment to a square  $m \times m$  image with column stride  $s$ ; Table 7 lists the values per length. Table 8 lists the per-(dataset, length) logit-normal mean  $P_{mean}$ ; the standard deviation is  $P_{std}=1.2$  throughout.

## E Baseline Implementations and Hyperparameters

Every baseline is run inside the same harness as GALA: each method’s generator and conditioning path follow its official architecture and released configuration, but data loading, train/validation splitting, checkpointing and generation evaluation are shared code, so all methods see identical data and are scored in the identical embedding space. Table 9 lists the settings that are common to all five baselines; the per-method settings follow in Table 10–Table 15.

### E.1 VerbalTS

VerbalTS (Gu et al., 2025) is an observation-space DDPM with multi-scale patch denoising. We keep its released text path exactly: frozen LongCLIP (zer0int/LongCLIP-GmP-ViT-L-14) per-token features, with no normalization or adapter inserted between the cached text features and VerbalTS’s own trainable text projector.

| Hyperparameter               | Value                                    |
|------------------------------|------------------------------------------|
| <i>Backbone (JiT/DiT-2D)</i> |                                          |
| Normalization / attention    | RMSNorm, QK-normalized                   |
| Positional encoding          | 2-D RoPE                                 |
| Feed-forward / conditioning  | SwiGLU, adaLN-Zero                       |
| Hidden size                  | 128                                      |
| Depth (layers)               | 4                                        |
| Heads (head dim)             | 4 (32)                                   |
| Patch size                   | 2                                        |
| Input channels               | 1                                        |
| Bottleneck dim               | 128                                      |
| Dropout (attn / proj)        | 0.1 / 0.1 (middle blocks)                |
| Register tokens              | 0                                        |
| Guidance / condition dropout | none / none                              |
| <i>Flow matching</i>         |                                          |
| Target                       | $X$ -prediction + velocity matching      |
| Timestep schedule            | logit-normal ( $P_{\text{std}}=1.2$ )    |
| $P_{\text{mean}}$            | per dataset (Table 8)                    |
| Noise scale $\sigma$         | 1.0                                      |
| <i>Sampling</i>              |                                          |
| Solver                       | Heun (deterministic ODE), 50 steps       |
| EMA decays                   | 0.9999 (ema1), 0.999 (ema2)              |
| <i>Optimization</i>          |                                          |
| Optimizer                    | AdamW ( $\beta=(0.9, 0.999)$ )           |
| Learning rate                | $1 \times 10^{-4}$ (constant, no warmup) |
| Weight decay                 | 0                                        |
| Gradient clipping            | 1.0                                      |
| Batch size                   | 512                                      |
| Epochs                       | 2000                                     |

Table 6: Stage-2 generator hyperparameters.

| Length $L$ | Image side $m$ | Stride $s$ | Image          |
|------------|----------------|------------|----------------|
| 24         | 6              | 4          | $6 \times 6$   |
| 48         | 8              | 6          | $8 \times 8$   |
| 96         | 10             | 10         | $10 \times 10$ |

 Table 7: Delay-embedding geometry per fragment length. Traffic at  $L=24$  uses  $m=8, s=8$  ( $8 \times 8$ ).

| Dataset     | $L=24$ | $L=48$ | $L=96$ |
|-------------|--------|--------|--------|
| ETTh1s      | 0.5    | 0.5    | 0.5    |
| ETTh1       | 0.8    | 0.5    | 0.0    |
| electricity | 0.5    | 0.5    | 0.5    |
| traffic     | -0.8   | -0.5   | -0.8   |

 Table 8: Per-(dataset, length) logit-normal timestep mean  $P_{\text{mean}}$ . The standard deviation is  $P_{\text{std}}=1.2$  in all settings.

| Setting                       | Value                      |
|-------------------------------|----------------------------|
| Benchmark                     | TSFragment-600K, 4 sources |
| Series normalization          | global min-max to $[0, 1]$ |
| Train / val split             | 90/10, split seed 0        |
| Val-loss seed                 | 12345                      |
| Gen-eval cadence              | every 100 epochs           |
| Gen-eval sample budget        | full validation split      |
| Metric embedder               | frozen reference aligner   |
| Metric seed / top- $K$ / bins | 123 / 5 / 32               |
| Tracked weight variants       | online, ema1, ema2         |
| Checkpoint selection          | lowest FID                 |
| Data workers                  | 8-16                       |

Table 9: Settings shared by all five baselines and by GALA. Checkpoint selection and the metric embedder follow the protocol in the Evaluation Protocol section, so no method is scored in its own embedding space.

GALA: Generation-Aware Cross-Modal Alignment for Text-to-Time-Series Synthesis

| Hyperparameter            | Value                        |
|---------------------------|------------------------------|
| <i>Text path</i>          |                              |
| Encoder                   | frozen LongCLIP-GmP-ViT-L-14 |
| Readout                   | per-token, $T=248$ , $D=768$ |
| Encoder dtype             | bfloat16                     |
| Pre-projector             | 768→512→64 (LN, LReLU)       |
| Text projector stages     | 3                            |
| <i>Denoiser</i>           |                              |
| Variates $n_{\text{var}}$ | 1                            |
| Model width $d$           | 64                           |
| Multi-patch scales $K$    | 3 (lengths 3, 6, 12)         |
| Residual blocks / heads   | 3 / 8                        |
| Diffusion-embedding dim   | 128                          |
| Variate / time side-emb.  | 16 / 112                     |
| Attention mask            | parallel (block-diagonal)    |
| Conditioning              | adaLN                        |
| <i>Diffusion</i>          |                              |
| Steps / schedule          | 50 / quad                    |
| $\beta$ range             | $10^{-4} \rightarrow 0.5$    |
| Sampler                   | DDIM, all 50 steps           |
| Prior noise scale         | 1.0                          |
| EMA decays                | 0.999 (ema1), 0.9999 (ema2)  |
| <i>Optimization</i>       |                              |
| Optimizer                 | AdamW                        |
| Learning rate / decay     | $1 \times 10^{-3}$ / 0       |
| Gradient clipping         | 1.0                          |
| Batch size / epochs       | 512 / 2000                   |

Table 10: VerbalTS baseline hyperparameters.

| Hyperparameter            | Value                          |
|---------------------------|--------------------------------|
| <i>Text path</i>          |                                |
| Encoder                   | frozen LongCLIP-GmP-ViT-L-14   |
| Readout                   | pooled CLS                     |
| Adapter                   | LayerNorm, hidden 512          |
| Harness condition dropout | 0 (disabled)                   |
| <i>Denoiser</i>           |                                |
| Input channels            | 1                              |
| Prototype dim / count     | 32 / 16                        |
| Latent unit               | 1                              |
| Model channels            | 32                             |
| Residual blocks per level | 2                              |
| Channel multipliers       | (1, 2, 4, 4)                   |
| Attention resolutions     | (1, 2, 4)                      |
| Attention heads           | 8                              |
| Transformer depth         | 1                              |
| Dropout                   | 0                              |
| Text-prototype fusion     | gated add                      |
| <i>Diffusion</i>          |                                |
| Steps / schedule          | 50 / quad                      |
| $\beta$ range             | $10^{-4} \rightarrow 0.5$      |
| Sampler / $\eta$          | DDIM (all steps) / 0.0         |
| Prototype drop (train)    | 0.5                            |
| Guidance scale (sampling) | 1.0                            |
| EMA decays                | 0.999 (ema1), 0.9999 (ema2)    |
| <i>Optimization</i>       |                                |
| Optimizer                 | AdamW                          |
| Learning rate / decay     | $1 \times 10^{-3}$ / $10^{-4}$ |
| Gradient clipping         | 1.0                            |
| Batch size / epochs       | 256 / 2000                     |

Table 11: BRIDGE baseline hyperparameters.

## E.2 BRIDGE

BRIDGE (Li et al., 2025) is an observation-space DDPM whose 1-D U-Net is cross-attended by domain prototypes extracted from an example series, with the pooled text embedding fused in by a gated conditioning MLP. Classifier-free guidance is BRIDGE’s own mechanism (dropping the prototype context to a learnable null embedding), so the harness-level condition dropout is disabled to avoid stacking two guidance mechanisms.

## E.3 T2S

T2S (Ge et al., 2025) is a two-stage latent method: a length-adaptive VAE is pretrained first, then a rectified-flow Transformer is trained in its latent space. Following the released implementation, conditioning uses the 128-d TextEmbedding column shipped with TSFragment-600K (OpenAI embeddings) rather than a text encoder trained here, and the latent geometry is fixed to the (64, 30) tensor the T2S Transformer expects.

## E.4 DiffuSETS

DiffuSETS (Lai et al., 2025) is a latent DDPM with a U-Net denoiser whose cross-attention reads the caption embedding. As with T2S, the conditioning signal is the dataset’s 128-d TextEmbedding column. One VAE is trained per (source, length) rather than across lengths. The guidance scale is kept at 1.0, i.e.

| Hyperparameter            | Value                       |
|---------------------------|-----------------------------|
| Text condition            | 128-d dataset TextEmbedding |
| Latent shape              | (64, 30), from the LA-VAE   |
| Generator                 | rectified-flow Transformer  |
| Train-time flow grid      | 100 steps                   |
| Sampling steps            | 10 (Euler)                  |
| Guidance scale            | 9.0                         |
| Condition dropout (train) | 0.3                         |
| EMA decays                | 0.9999 (ema1), 0.999 (ema2) |
| Optimizer                 | AdamW                       |
| Learning rate / decay     | $1 \times 10^{-4}$ / 0      |
| Gradient clipping         | 1.0                         |
| Batch size / epochs       | 256 / 2000                  |

Table 12: T2S baseline hyperparameters (stage 2). The stage-1 LA-VAE is in Table 15.

| Hyperparameter                | Value                          |
|-------------------------------|--------------------------------|
| Text condition                | 128-d dataset TextEmbedding    |
| Latent shape                  | (4, $L/2$ ), from the VAE      |
| U-Net levels / kernel / heads | 3 / 3 / 8                      |
| Diffusion steps / schedule    | 50 / quad                      |
| $\beta$ range                 | $10^{-4} \rightarrow 0.5$      |
| Sampler                       | DDIM                           |
| Guidance scale                | 1.0 (none)                     |
| Condition dropout (train)     | 0                              |
| EMA decays                    | 0.9999 (ema1), 0.999 (ema2)    |
| Optimizer                     | AdamW                          |
| Learning rate / decay         | $1 \times 10^{-3}$ / $10^{-4}$ |
| Gradient clipping             | 1.0                            |
| Batch size / epochs           | 256 / 2000                     |

Table 13: DiffuSETS baseline hyperparameters (stage 2). The stage-1 VAE is in Table 15.

no classifier-free guidance, which is faithful to the released setting; condition dropout is correspondingly 0.

## E.5 Text2Motion

Text2Motion (Guo et al., 2022) is adapted from text-to-motion synthesis: a movement autoencoder is pretrained per (source, length), and a text-conditioned autoregressive CVAE then generates in its latent space. The optimizer settings follow the released Comp trainer (plain Adam, gradient clipping 0.5, no weight decay); conditioning again uses the dataset’s 128-d TextEmbedding column.

## E.6 Stage-1 Pretraining for the Two-Stage Baselines

T2S, DiffuSETS and Text2Motion each require a pretrained autoencoder before their generator can be trained. In all three cases the stage-1 model uses the same validation ratio and split seed as its stage-2 generator, so the autoencoder never sees a held-out fragment. Table 15 lists the settings.

# F Evaluation Protocol

## F.1 The Evaluation Aligner

Fidelity is measured by FID and caption adherence by CTPP ( $\uparrow$ ) and JFTSD ( $\downarrow$ ), computed with our reimplementation of the ConTSG-Bench metrics (Lan et al., 2026). All three live in the embedding space of a single text–time-series dual encoder (the evaluation aligner) that exists only for measurement. FID is the Fréchet distance between the embeddings of real and generated fragments, CTPP the mean cosine similarity between a caption embedding and the embedding of the fragment generated from that caption, and JFTSD the Fréchet distance over the concatenated [series, text] embeddings.

Because GALA is itself a text–time-series alignment, an evaluator built from the same components might favor it by construction. We therefore build the evaluation aligner from disjoint parts. Its text tower is BGE-large-en-v1.5 (Xiao et al., 2024), a different model family from the Embedding-Gemma-300M that GALA adapts; its series tower is a convolutional encoder trained from scratch, not initialized from Chronos-2 or any other time-series foundation model. No weights are shared with either GALA tower in either direction. We do not claim that the two text encoders were pretrained on disjoint web corpora, since neither model’s pretraining data is public at that granularity; we claim only that the evaluator reuses no parameters from the models it scores.

One evaluation aligner is fit per (source, length) configuration on that configuration’s training split, under exactly the split used to train every generator it scores (10% held out, split seed 0); we verified

| Hyperparameter                 | Value                       |
|--------------------------------|-----------------------------|
| Text condition                 | 128-d dataset TextEmbedding |
| Text latent dim                | 512                         |
| Attention vector dim           | 512                         |
| Latent dim $z$                 | 128                         |
| Prior / post. / dec. hidden    | 1024 / 1024 / 1024          |
| Prior / post. / dec. layers    | 1 / 1 / 1                   |
| Recon. weights (series / mov.) | 1.0 / 1.0                   |
| KL weight                      | 0.005                       |
| Teacher-forcing ratio          | 0.4                         |
| EMA decays                     | 0.9999 (ema1), 0.999 (ema2) |
| Optimizer                      | Adam                        |
| Learning rate / decay          | $2 \times 10^{-4}$ / 0      |
| Gradient clipping              | 0.5                         |
| Batch size / epochs            | 32 / 2000                   |

Table 14: Text2Motion baseline hyperparameters (stage 2). The stage-1 movement autoencoder is in Table 15.

| Hyperparameter                           | Value                          |
|------------------------------------------|--------------------------------|
| <i>T2S: length-adaptive VAE</i>          |                                |
| Training lengths                         | mixed {24, 48, 96}             |
| Block hidden / residual layers           | 128 / 2                        |
| Residual hidden / embedding dim          | 256 / 64                       |
| Learning rate                            | $1 \times 10^{-3}$             |
| Batch size / epochs / seed               | 512 / 2000 / 42                |
| <i>DiffuSETS: convolutional VAE</i>      |                                |
| Training lengths                         | one model per length           |
| Downsample factor                        | 2                              |
| Latent / base channels                   | 4 / 128                        |
| KL weight (constant)                     | 0.01                           |
| Optimizer                                | AdamW                          |
| Learning rate / decay                    | $1 \times 10^{-3}$ / $10^{-4}$ |
| Gradient clipping                        | 1.0                            |
| Batch size / epochs / seed               | 256 / 1000 / 42                |
| <i>Text2Motion: movement autoencoder</i> |                                |
| Training lengths                         | one model per length           |
| Movement latent dim                      | 512                            |
| Enc. / dec. hidden                       | 512 / 512                      |
| Sparsity / smoothness weights            | 0.001 / 0.001                  |
| Optimizer                                | Adam (no clipping)             |
| Learning rate / decay                    | $1 \times 10^{-4}$ / 0         |
| Batch size / epochs / seed               | 128 / 270 / 42                 |

Table 15: Stage-1 autoencoder pretraining for the two-stage baselines. Each stage-1 model reuses its stage-2 validation ratio (0.1) and split seed (0).

that the held-out fragments on which metrics are computed do not occur in the aligner’s own training data. It is then frozen and reused unchanged for every method, ablation, and figure in this paper, so all reported numbers are commensurable and no model is scored in a space it helped produce.

## F.2 Checkpoint Selection and Aggregation

Generation metrics are evaluated every 100 epochs for all three model variants (online, ema1, ema2). For each run we select the single checkpoint (epoch  $\times$  variant) with the lowest FID on the full held-out split, then re-sample that checkpoint 3 times with independent sampling seeds (100/200/300) and report mean  $\pm$  sample standard deviation over those 3 draws. Both stages are trained once with a fixed seed (0), so the reported variation is sampling variance: it quantifies how much a reported number moves when the same trained model is re-sampled, and does not capture run-to-run training variance. We state this explicitly because the comparisons in Table 1 of the main paper turn on small absolute differences; Appendix G quantifies that variance and tests the resulting ordering.

## G Variability and Significance Testing

Table 1 of the main paper reports means only, and several columns are decided by differences in the third decimal. This appendix reports the corresponding dispersion and tests whether the resulting ordering is statistically supported.

### G.1 Per-Cell Sampling Variability

Table 17 gives the sample standard deviation over the 3 sampling seeds for every one of the 216 cells of Table 1 (6 methods  $\times$  4 datasets  $\times$  3 lengths  $\times$  3 metrics). Dispersion is small relative to the reported gaps:



the largest standard deviation anywhere is 0.0165 (DiffuSETS, FID on ETTh1s-96), the largest for GALA is 0.0033, 98.1% of all cells are below 0.01, and the median cell is 0.0011 (FID), 0.0026 (CTTP), 0.0024 (JFTSD). As a per-cell check against the strongest baseline we ran a Welch  $t$ -test (GALA vs. VerbalTS,  $n = 3$  per side) in each of the 36 dataset  $\times$  length  $\times$  metric cells and corrected the 36  $p$ -values with Holm’s procedure: 29 of the 30 cells we win remain significant at  $\alpha = 0.05$ , and across those winning cells the median gap is  $34\times$  the pooled standard deviation. The three cells that are not resolved are traffic-48 JFTSD (a win) and the two FID cells of ETTh1s-96 and ETTm1-96 (both losses), where the two methods are within one standard deviation of each other. As noted above, this variance is sampling-only; it bounds the noise in each reported number but not the variability of retraining.

## G.2 Wilcoxon Signed-Rank Tests Against the Baselines

Following the standard protocol for comparing methods over multiple problems (Demšar, 2006), we test GALA against each baseline with a one-sided Wilcoxon signed-rank test, Holm-corrected over the five comparisons. We run these per metric, so that the 12 paired observations entering a test (four datasets  $\times$  three lengths) are all on the same scale; a signed-rank test that mixed FID, CTTP, and JFTSD in one sample would weight the metrics by their arbitrary numeric ranges. Table 16 reports one-sided  $p$ -values.

The outcome is uniform against four of the five baselines: GALA wins 12/12 settings on essentially every metric and every Holm-corrected  $p$  is below 0.01. Against VerbalTS the picture is split. On both caption-adherence metrics the advantage is significant (CTTP 12/12, JFTSD 11/12, both  $p = 1.2 \times 10^{-3}$ ), whereas on FID it is not (7/12 wins,  $p = 0.13$ ): the two methods are statistically indistinguishable on fidelity. We therefore do not claim a fidelity advantage over VerbalTS, only parity, and locate our gain on caption adherence.

For completeness we also ran the test in the form suggested by the layout of Table 1, i.e. on the 12 columns of a single length taken together, converting each column to the relative improvement of GALA over VerbalTS so that the three metric scales are commensurate. This gives  $p = 7 \times 10^{-4}$  at  $L = 24$  (11/12 columns won, median relative gain 30.3%),  $p = 7 \times 10^{-4}$  at  $L = 48$  (11/12, 37.3%),  $p = 0.12$  at  $L = 96$  (8/12, 12.8%), and  $p = 3.7 \times 10^{-6}$  on all 36 columns pooled (30/36, 28.6%).

A per-length test of this kind pools the three metrics, so it inherits the FID parity reported above, and that is what the  $L = 96$  value reflects rather than any weakening of caption adherence. Reading the same results one (dataset, length) pair at a time confirms this. A signed-rank test cannot be run at that granularity, because each pair contributes only three columns, so the smallest attainable one-sided  $p$  is 0.125; we therefore use the per-cell Welch tests of the previous subsection. GALA wins *both* caption-adherence metrics in 11 of the 12 pairs, and all three metrics in 7 of them; five of the six lost cells are FID, the sixth being JFTSD on traffic-96, the only pair in which an adherence metric goes against us. By metric, GALA wins CTTP on 4/4 domains at *every* length and JFTSD on 4/4, 4/4 and 3/4 domains at  $L = 24/48/96$ , whereas FID goes 3/4, 3/4 and 1/4. The caption-adherence advantage therefore holds at  $L = 96$  much as it does at the two shorter lengths; what narrows at  $L = 96$  is fidelity, on which we claim only parity in the first place.

## H Morphological Diversity of the Benchmark Sources

Traffic is where GALA’s margin is thinnest: three of the six cells it loses to VerbalTS sit there, and traffic-96 is the only (dataset, length) pair in which a caption-adherence metric goes against us. This appendix

| Ours vs.    | FID  |                   | CTTP |                   | JFTSD |                   |
|-------------|------|-------------------|------|-------------------|-------|-------------------|
|             | W-L  | $p_{\text{Holm}}$ | W-L  | $p_{\text{Holm}}$ | W-L   | $p_{\text{Holm}}$ |
| VerbalTS    | 7-5  | 0.133             | 12-0 | 0.0012            | 11-1  | 0.0012            |
| Text2Motion | 12-0 | 0.0012            | 12-0 | 0.0012            | 12-0  | 0.0012            |
| BRIDGE      | 11-1 | 0.0068            | 12-0 | 0.0012            | 12-0  | 0.0012            |
| DiffuSETS   | 12-0 | 0.0012            | 12-0 | 0.0012            | 12-0  | 0.0012            |
| T2S         | 12-0 | 0.0012            | 12-0 | 0.0012            | 12-0  | 0.0012            |

Table 16: Pairwise one-sided Wilcoxon signed-rank tests of GALA against each baseline, run separately per metric over the 12 settings (four datasets  $\times$  three lengths), with Holm correction over the five comparisons within a metric. W-L counts the settings won and lost. The only comparison that fails to reach significance is FID against VerbalTS, i.e. the two are indistinguishable on fidelity.

| Method                            | ETTh1s |      |       | ETTm1 |      |       | Electricity |      |       | Traffic |      |       |
|-----------------------------------|--------|------|-------|-------|------|-------|-------------|------|-------|---------|------|-------|
|                                   | FID    | CTTP | JFTSD | FID   | CTTP | JFTSD | FID         | CTTP | JFTSD | FID     | CTTP | JFTSD |
| <i>Length <math>L = 24</math></i> |        |      |       |       |      |       |             |      |       |         |      |       |
| DiffuSETS                         | 3.4    | 4.4  | 5.4   | 1.6   | 1.1  | 2.3   | 3.3         | 4.4  | 3.6   | 1.7     | 1.0  | 2.6   |
| T2S                               | 1.0    | 6.6  | 6.3   | 1.9   | 0.7  | 3.2   | 2.6         | 4.7  | 2.8   | 6.1     | 1.3  | 4.3   |
| BRIDGE                            | 0.5    | 8.2  | 8.0   | 1.5   | 2.3  | 2.6   | 1.9         | 3.2  | 3.8   | 9.1     | 3.7  | 7.3   |
| Text2Motion                       | 0.9    | 2.1  | 1.2   | 0.6   | 3.0  | 2.4   | 1.5         | 3.0  | 2.1   | 0.4     | 3.9  | 1.7   |
| VerbalTS                          | 0.4    | 5.4  | 3.9   | 0.1   | 0.3  | 0.5   | 0.6         | 2.6  | 1.4   | 1.5     | 4.1  | 2.6   |
| <b>Ours</b>                       | 0.2    | 1.4  | 0.4   | 0.1   | 0.5  | 0.1   | 0.2         | 2.3  | 0.5   | 0.2     | 0.3  | 0.2   |
| <i>Length <math>L = 48</math></i> |        |      |       |       |      |       |             |      |       |         |      |       |
| DiffuSETS                         | 1.3    | 8.8  | 7.7   | 1.2   | 1.0  | 2.0   | 1.6         | 2.1  | 2.5   | 0.9     | 3.2  | 4.1   |
| T2S                               | 0.6    | 3.7  | 3.3   | 1.9   | 0.2  | 2.2   | 1.7         | 2.8  | 2.9   | 0.7     | 3.5  | 2.5   |
| BRIDGE                            | 1.0    | 6.5  | 6.5   | 1.4   | 1.7  | 1.6   | 1.3         | 2.6  | 2.9   | 4.7     | 6.3  | 7.1   |
| Text2Motion                       | 0.7    | 4.2  | 3.6   | 0.8   | 2.4  | 2.5   | 1.0         | 1.5  | 1.7   | 1.5     | 2.2  | 1.8   |
| VerbalTS                          | 0.9    | 3.8  | 5.5   | 0.2   | 2.4  | 0.7   | 0.8         | 2.8  | 1.4   | 1.4     | 3.0  | 1.9   |
| <b>Ours</b>                       | 0.1    | 3.3  | 1.4   | 0.1   | 0.3  | 0.2   | 1.0         | 1.3  | 0.8   | 0.3     | 1.2  | 0.6   |
| <i>Length <math>L = 96</math></i> |        |      |       |       |      |       |             |      |       |         |      |       |
| DiffuSETS                         | 16.5   | 2.8  | 10.2  | 2.1   | 1.1  | 2.4   | 6.7         | 3.7  | 8.1   | 12.8    | 2.0  | 13.0  |
| T2S                               | 0.6    | 7.5  | 8.8   | 0.5   | 1.4  | 1.4   | 1.8         | 1.3  | 2.2   | 8.2     | 1.1  | 5.2   |
| BRIDGE                            | 3.6    | 5.0  | 4.0   | 0.9   | 1.9  | 0.3   | 1.9         | 4.1  | 1.7   | 5.0     | 3.8  | 1.7   |
| Text2Motion                       | 3.0    | 4.1  | 6.3   | 1.5   | 0.7  | 1.4   | 1.5         | 4.4  | 3.1   | 6.7     | 5.5  | 6.2   |
| VerbalTS                          | 1.3    | 8.1  | 5.0   | 0.3   | 1.7  | 0.6   | 0.8         | 1.7  | 1.6   | 0.5     | 3.0  | 0.3   |
| <b>Ours</b>                       | 0.7    | 1.5  | 0.8   | 0.7   | 0.7  | 0.5   | 0.4         | 0.7  | 0.3   | 1.1     | 1.8  | 1.3   |

Table 17: Sample standard deviation over the 3 sampling seeds for every cell of Table 1 in the main paper. **All values are  $\times 10^{-3}$** , so an entry of 3.4 means 0.0034. The largest entry in the whole table is 16.5 (0.0165; DiffuSETS, FID on ETTh1s-96) and the largest for GALA is 3.3 (0.0033).

characterizes why, by measuring how many distinct *shapes* each source actually contains.

**Protocol** Sources differ by orders of magnitude in absolute scale, so we first  $z$ -normalize every fragment individually; this removes level and amplitude and leaves only morphology. On 2000 fragments per (source, length) we then compute the PCA spectrum of the shape population and a  $k=6$  KMeans clustering, and report: the share of shape variance in the first two components ( $\text{PCA}_2$ ); the number of components needed to reach 90% ( $\text{PC}_{90}$ ); and the participation ratio ( $\sum_i \lambda_i^2 / \sum_i \lambda_i^2$  of the spectrum, an effective number of shape dimensions (eff. rank). A source with few distinct shapes has high  $\text{PCA}_2$  and low  $\text{PC}_{90}$  and eff. rank.

**Traffic is an extreme outlier** Table 18 shows the result, and the ordering is the same at all three lengths. Traffic needs only 2–4 principal components to explain 90% of its shape variance, against 6–11 for the other three sources, and its effective rank (2.3–2.6) is roughly half of theirs. Its KMeans clusters are also the tightest and best separated (silhouette 0.35–0.43 versus 0.15–0.25 elsewhere), and random fragment pairs are far more correlated (mean  $|r|$  0.55–0.59 versus 0.31–0.45). Figure 6 makes the reason visible: the six Traffic cluster centroids are the same daily template read at six different phases, whereas the ETTh1s centroids are genuinely different morphologies. Traffic is, in effect, a one-waveform benchmark.

**How this relates to our results, and what it does not show** A source with essentially one shape family leaves a caption little morphology to specify, so there is less for a text–series alignment to exploit; a narrow real distribution also makes fidelity easy to reach by generating near the mode. Consistent with that, averaged over the three lengths our relative gain over VerbalTS on traffic is +21.7% on CTTP, +0.1% on JFTSD and −11.1% on FID, against +30.8%, +54.5% and +20.7% on the other three sources.

| Source      | $L = 24$         |                  |            | $L = 48$         |                  |            | $L = 96$         |                  |            |
|-------------|------------------|------------------|------------|------------------|------------------|------------|------------------|------------------|------------|
|             | PCA <sub>2</sub> | PC <sub>90</sub> | eff.       | PCA <sub>2</sub> | PC <sub>90</sub> | eff.       | PCA <sub>2</sub> | PC <sub>90</sub> | eff.       |
| ETTh1s      | 0.62             | 8                | 4.6        | 0.47             | 10               | 6.4        | 0.56             | 9                | 4.7        |
| ETTm1       | 0.64             | 9                | 3.6        | 0.66             | 11               | 3.8        | 0.62             | 11               | 4.7        |
| Electricity | 0.71             | 6                | 3.8        | 0.54             | 8                | 5.5        | 0.44             | 11               | 7.2        |
| Traffic     | <b>0.93</b>      | <b>2</b>         | <b>2.3</b> | <b>0.90</b>      | <b>2</b>         | <b>2.5</b> | <b>0.87</b>      | <b>4</b>         | <b>2.6</b> |

Table 18: Shape diversity of the four sources, measured on 2000 per-series  $z$ -normalized fragments per configuration. PCA<sub>2</sub> is the share of shape variance in the first two principal components, PC<sub>90</sub> the number of components reaching 90%, and eff. the participation ratio of the spectrum. Traffic (bold) needs the fewest dimensions at every length, i.e. it contains the fewest distinct shapes.

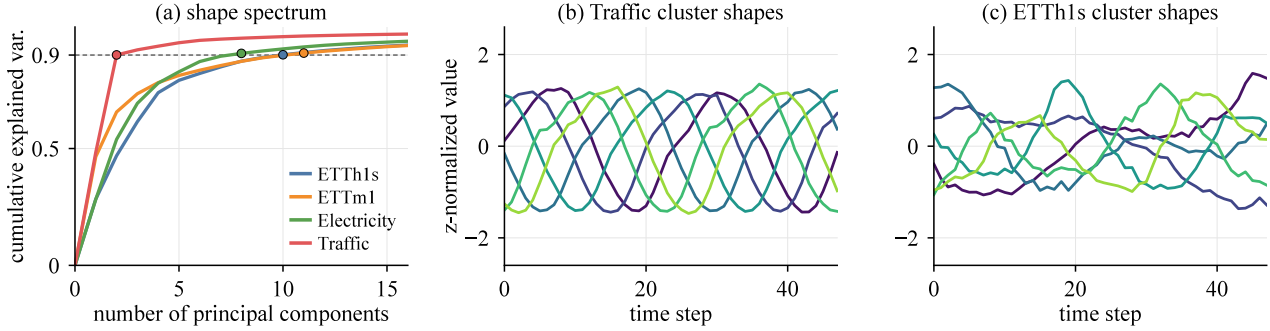


Figure 6: Morphological diversity at  $L = 48$ . (a) Cumulative PCA explained variance of the  $z$ -normalized fragments; markers sit where each source reaches 90% (dashed line). Traffic saturates after two components while the others need eight to eleven. (b, c) The six KMeans centroid shapes of Traffic and of ETTh1s on identical axes: Traffic’s clusters are one daily template at six different phases, whereas ETTh1s’s are distinct morphologies.

We deliberately stop short of a causal claim. Across the 12 (source, length) configurations the rank correlation between effective shape rank and our relative gain is not significant for any metric (Spearman  $\rho = +0.01$  for CTP,  $+0.29$  for JFTSD,  $+0.34$  for FID; all  $p > 0.25$ ), which four sources are far too few to resolve. We therefore report the diversity measurement as a characterization of the benchmark, in that traffic is quantitatively degenerate in morphology at every length, and as the most plausible reading of where our method has the least room to help, not as a demonstrated mechanism.

## I Sweeps

We report the following sweeps in the main paper: the auxiliary-loss weight  $\lambda_{dw} \in \{0, 1, 2, 4, 8\}$ ; the fine-tuning method  $\in \{\text{LoRA}, \text{FT-Full}, \text{FT-Head}\}$ ; the adapted tower  $\in \{\text{text-only}, \text{series-only}, \text{both}\}$ ; and the series normalization  $\in \{\text{global } z\text{-score}, \text{per-sample}\}$ . In every sweep, the cell that coincides with our default configuration reports the same numbers as the corresponding entry of Table 1 in the main paper.

## J Software and Hardware

All experiments run on a single NVIDIA RTX PRO 6000 Blackwell GPU. Our implementation uses Python 3.11, PyTorch 2.11.0, HuggingFace Transformers 4.57.6, PEFT 0.18.1, chronos-forecasting 2.2.2, Accelerate 1.12.0, and NumPy 2.1.3.